Measurement Design - Photoelectric Effect

Date: 2026.10.19, 2026.10.26

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Task 2:

The geometric parameters of the setup:

Distance between the laser and the camera: 20cm

Distance between the light source and the first difraction orders: 7. 45 cm

```
warning('off', 'all')
l = 19.75;
lambda = 532e-9;
m = 1;

X_test= 7.05;

angle = atan(X_test/l);
d = lambda / sin(angle) * 1e6
```

```
d =
1.5825
```

We calculated a difraction grating width of 1.5825 nanometers. The result might not be precise, since we were not able to measure the distances down to a nanometer precission, part of this was caused by the 0.5 millimeter uncertainity of the measuring tape and the ruller.

Task 4:

```
X_uv = 5.15;
angle = atan(X_uv/l);
lambda_uv = sin(angle) * d
```

```
lambda_uv =
0.3993
```

```
X_uv_minus = X_uv - 0.5;
X_uv_plus = X_uv + 0.5;
lambda_uv_minus = sin(atan(X_uv_minus/l)) * d;
lambda_uv_plus = sin(atan(X_uv_plus/l)) * d;
error_dif_uv = diff([lambda_uv_minus, lambda_uv_plus])
```

```
error_dif_uv =
0.0726
```

```
X_blue = 6.1;
angle = atan(X_blue/l);
lambda_blue = sin(angle) * d
```

```
lambda_blue =
0.4670
```

```
X_blue_minus = X_blue - 0.5;
X_blue_plus = X_blue + 0.5;
lambda_blue_minus = sin(atan(X_blue_minus/l)) * d;
lambda_blue_plus = sin(atan(X_blue_plus/l)) * d;
error_dif_blue = diff([lambda_blue_minus, lambda_blue_plus])
```

```
error_dif_blue =
0.0699
```

```
X_green = 6.9;
angle = atan(X_green/l);
lambda_green = sin(angle) * d
```

```
lambda_green =
0.5219
```

```
X_green_minus = X_green - 0.5;
X_green_plus = X_green + 0.5;
lambda_green_minus = sin(atan(X_green_minus/l)) * d;
lambda_green_plus = sin(atan(X_green_plus/l)) * d;
error_dif_green = diff([lambda_green_minus, lambda_green_plus])
```

```
error_dif_green =
0.0674
```

```
X_yellow = 7.8;
angle = atan(X_yellow/l);
lambda_yellow = sin(angle) * d
```

```
lambda_yellow =
0.5813
```

```
X_yellow_minus = X_yellow - 0.5;
X_yellow_plus = X_yellow + 0.5;
lambda_yellow_minus = sin(atan(X_yellow_minus/l)) * d;
lambda_yellow_plus = sin(atan(X_yellow_plus/l)) * d;
error_dif_yellow = diff([lambda_yellow_minus, lambda_yellow_plus])
```

```
error_dif_yellow =
0.0645
```

```
X_red = 8.4;
```

```
angle = atan(X_red/l);
lambda_red = sin(angle) * d
```

```
lambda_red =
0.6194
```

```
X_red_minus = X_red - 0.5;
X_red_plus = X_red + 0.5;
lambda_red_minus = sin(atan(X_red_minus/l)) * d;
lambda_red_plus = sin(atan(X_red_plus/l)) * d;
error_dif_red = diff([lambda_red_minus, lambda_red_plus])
```

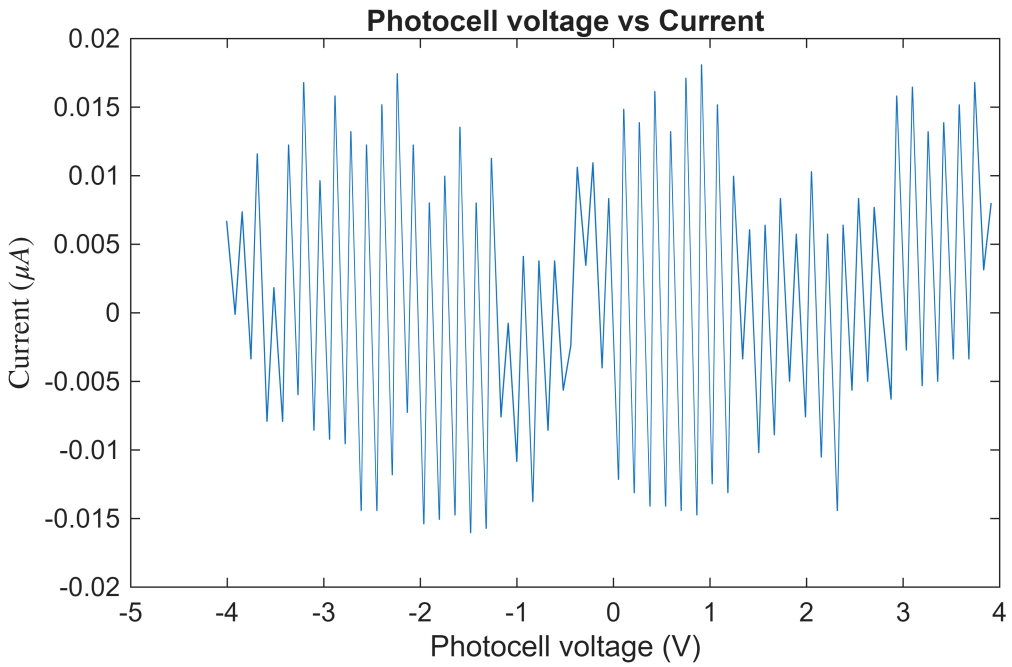
```
error_dif_red =
0.0624
```

The wavelength of for example the blue LED was calculated to be around 467 nanometers. This result is only an approximation, since the difraction grating apeared on the screen of the camera as a circle, and it was not possible to precisely measure the center of it, so there might be a slight misapproximation because of it.

Task 6:

The red and the yellow LEDs give off a much smaller current, since they have a lot smaller energy than the others.

```
data = readtable("dark.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
xlabel("Photocell voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
title("Photocell voltage vs Current");
```




```
difference = diff(intensity);
noise = mean(difference)*1e6
```

```
noise =
13.2644
```

```
figure;
hold on;
data = readtable("blue.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity, "Color","blue");

data = readtable("uv.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity, "Color","#8F00FF");

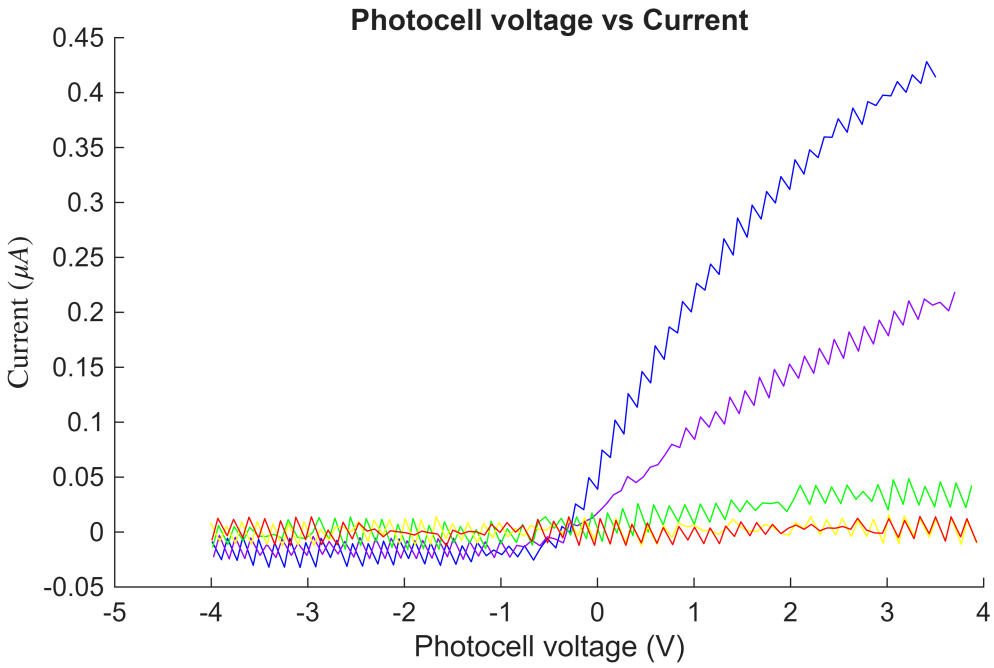
data = readtable("green.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity, "Color","green");

data = readtable("yellow.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity, "Color","yellow");

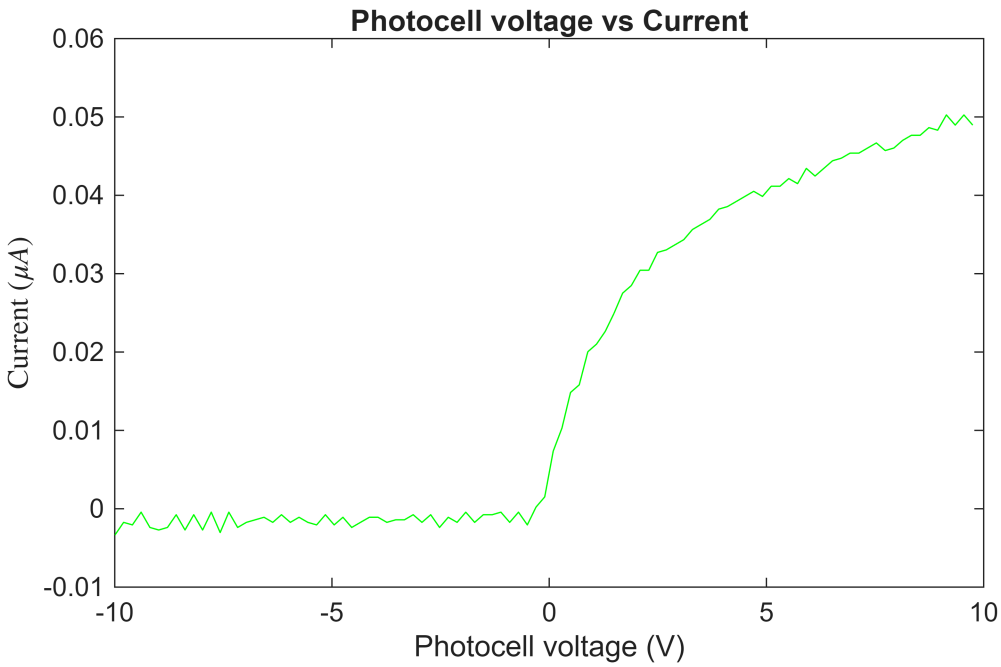
data = readtable("red.txt");
```

```
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity, "Color","red");

xlabel("Photocell voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
title("Photocell voltage vs Current")
```




```
figure;
data = readtable("green_wide.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity, "Color","green");
xlabel("Photocell voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
title("Photocell voltage vs Current");
```



Task 7:

```
Uth_arr = zeros(5,1)
```

```
Uth_arr = 5×1
     0
     0
     0
     0
     0
```

```
wavelengths = [0.3993,0.4670,0.5219,0.5813,0.6194];
c = 2.9992458 * 1e8* 1e6;
Freq_arr = c ./ wavelengths
```

```
Freq_arr = 1×5
10^14 ×
    7.5113    6.4224    5.7468    5.1595    4.8422
```

```
figure;
hold on;
data = readtable("uv.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position < -.5);
constantPosition = position(position < -.5);
constFit = polyfit(constantPosition,constantIntensity,1);
```

```
constSlope = constFit(1)
```

```
constSlope =
5.3375e-04
```

```
constIntercept = constFit(2)
```

```
constIntercept =
-0.0133
```

```
x = linspace(-4,4,100);
plot(x,constIntercept + x * constSlope);

linIntensity = intensity(position > 0 & position < 1);
linPosition = position(position > 0 & position < 1);
linFit = polyfit(linPosition, linIntensity, 1);
linSlope = linFit(1)
```

```
linSlope =
0.0746
```

```
linIntercept = linFit(2)
```

```
linIntercept =
0.0190
```

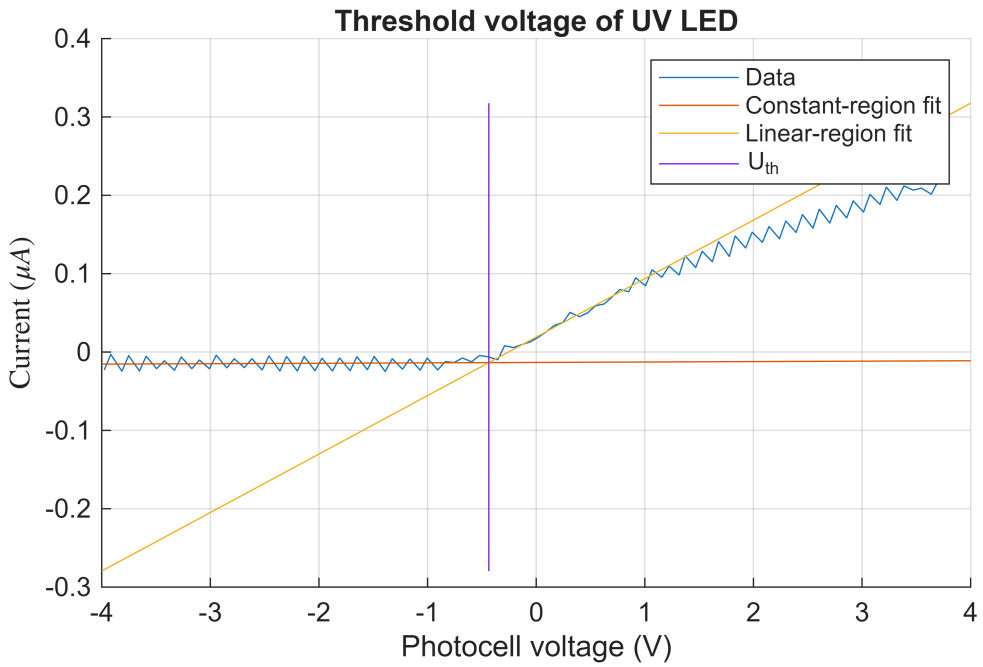
```
plot(x, linIntercept + linSlope * x);

Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
-0.4363
```

```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(linIntercept + -4*linSlope, linIntercept +
4*linSlope,20))

xlabel("Photocell voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{th}");
title("Threshold voltage of UV LED")
grid on;
```




```
Uth_arr(1) = Uth;
```

```
figure;
hold on;
data = readtable("blue.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position < -.5);
constantPosition = position(position < -.5);
constFit = polyfit(constantPosition,constantIntensity,1);
constSlope = constFit(1)
```

```
constSlope =
3.3641e-05
```

```
constIntercept = constFit(2)
```

```
constIntercept =
-0.0180
```

```
x = linspace(-4,4,100);
plot(x,constIntercept + x * constSlope);

linIntensity = intensity(position > 0 & position < 1);
linPosition = position(position > 0 & position < 1);
linFit = polyfit(linPosition, linIntensity, 1);
linSlope = linFit(1)
```

```
linSlope =
0.1608
```

```
linIntercept = linFit(2)
```

```
linIntercept =
0.0572
```

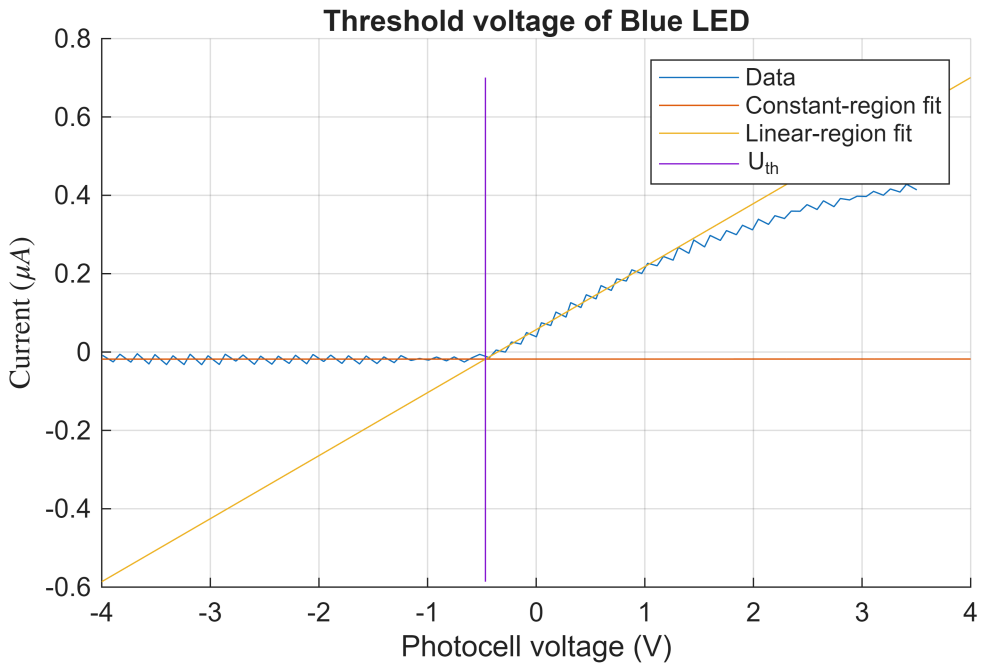
```
plot(x, linIntercept + linSlope * x);

Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
-0.4674
```

```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(linIntercept + -4*linSlope, linIntercept +
4*linSlope,20))

xlabel("Photocell voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{th}");
title("Threshold voltage of Blue LED")
grid on;
```




```
Uth_arr(2) = Uth;
```

```
figure;
hold on;
data = readtable("green.txt");
intensity = data{:, 2};
```

```
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position < -.5);
constantPosition = position(position < -.5);
constFit = polyfit(constantPosition,constantIntensity,1);
constSlope = constFit(1)
```

```
constSlope =
4.3016e-04
```

```
constIntercept = constFit(2)
```

```
constIntercept =
-3.2807e-04
```

```
x = linspace(-4,4,100);
plot(x,constIntercept + x * constSlope);

linIntensity = intensity(position > 0 & position < 1);
linPosition = position(position > 0 & position < 1);
linFit = polyfit(linPosition, linIntensity, 1);
linSlope = linFit(1)
```

```
linSlope =
0.0184
```

```
linIntercept = linFit(2)
```

```
linIntercept =
0.0030
```

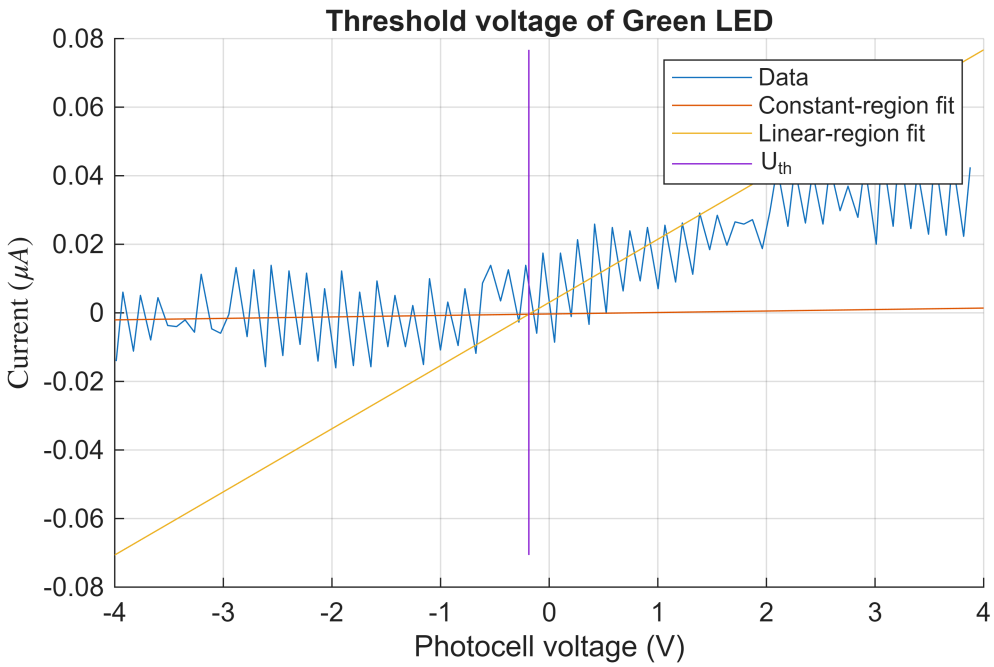
```
plot(x, linIntercept + linSlope * x);

Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
-0.1873
```

```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(linIntercept + -4*linSlope, linIntercept +
4*linSlope,20))

xlabel("Photocell voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{th}");
title("Threshold voltage of Green LED")
grid on;
```




```
Uth_arr(3) = Uth;
```

```
figure;
hold on;
data = readtable("yellow_wide.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position > -7.5 & position < -2.5);
constantPosition = position(position > -7.5 & position < -2.5);
constFit = polyfit(constantPosition,constantIntensity,1);
constSlope = constFit(1)
```

```
constSlope =
2.7996e-06
```

```
constIntercept = constFit(2)
```

```
constIntercept =
6.5592e-04
```

```
x = linspace(-10,10,100);
plot(x,constIntercept + x * constSlope);

linIntensity = intensity(position > 0 & position < 5);
linPosition = position(position > 0 & position < 5);
linFit = polyfit(linPosition, linIntensity, 1);
linSlope = linFit(1)
```

```
linSlope =
1.8787e-04
```

```
linIntercept = linFit(2)
```

```
linIntercept =
9.2184e-04
```

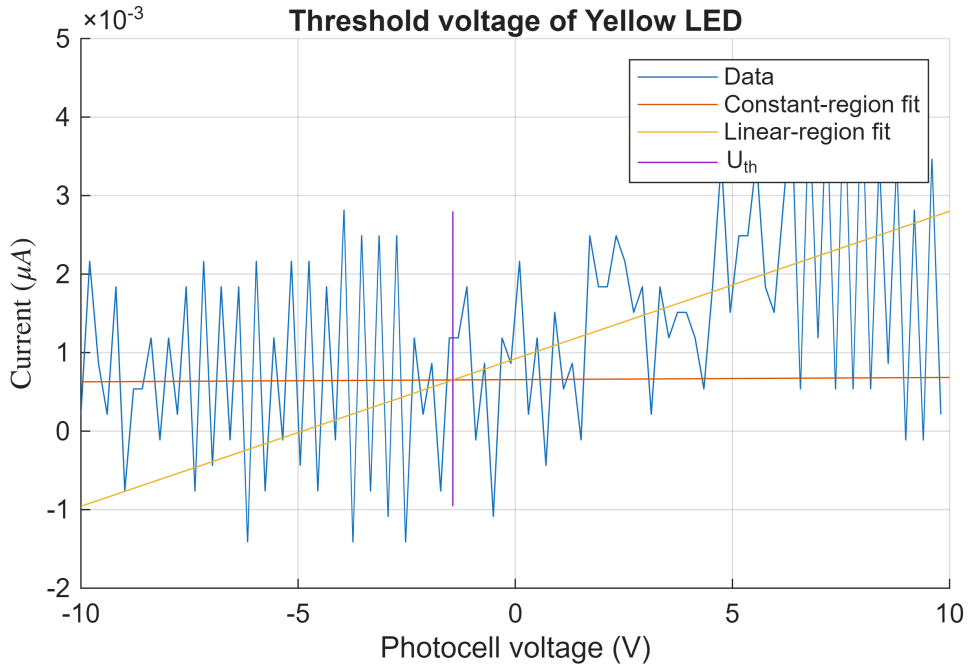
```
plot(x, linIntercept + linSlope * x);

Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
-1.4369
```

```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(linIntercept + -10*linSlope, linIntercept +
10*linSlope,20))

xlabel("Photocell voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{th}");
title("Threshold voltage of Yellow LED")
grid on;
```




```
Uth_arr(4) = Uth;
```

```
figure;
hold on;
data = readtable("red_wide.txt");
intensity = data{:, 2};
```

```
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position < -2);
constantPosition = position(position < -2);
constFit = polyfit(constantPosition,constantIntensity,1);
constSlope = constFit(1)
```

```
constSlope =
3.3626e-05
```

```
constIntercept = constFit(2)
```

```
constIntercept =
7.9064e-04
```

```
x = linspace(-10,10,100);
plot(x,constIntercept + x * constSlope);

linIntensity = intensity(position > 1 & position < 7);
linPosition = position(position > 1 & position < 7);
linFit = polyfit(linPosition, linIntensity, 1);
linSlope = linFit(1)
```

```
linSlope =
1.6809e-04
```

```
linIntercept = linFit(2)
```

```
linIntercept =
8.7737e-04
```

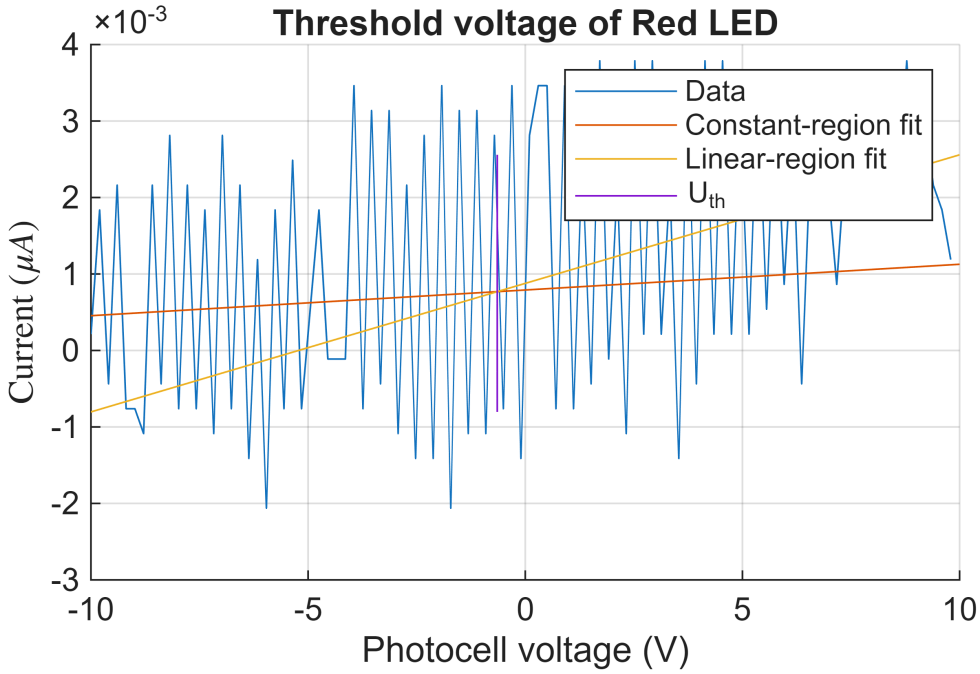
```
plot(x, linIntercept + linSlope * x);

Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
-0.6450
```

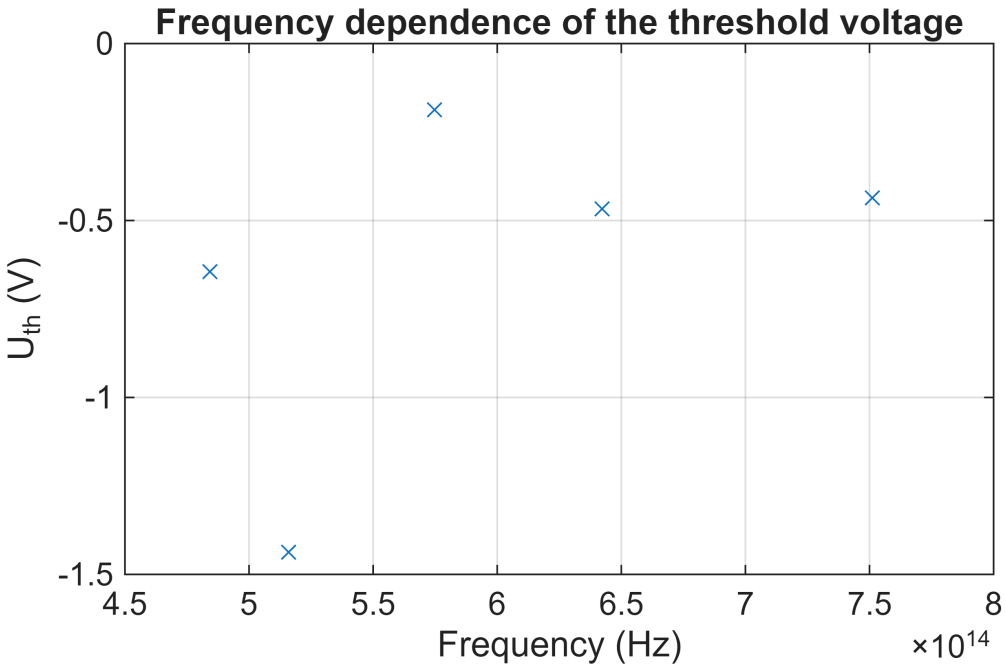
```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(linIntercept + -10*linSlope, linIntercept +
10*linSlope,20))

xlabel("Photocell voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{th}");
title("Threshold voltage of Red LED")
grid on;
```




```
Uth_arr(5) = Uth;
```

```
figure;
plot(Freq_arr,Uth_arr,'LineStyle','none','Marker','x')
ylabel("U_{th} (V)")
xlabel("Frequency (Hz)")
title("Frequency dependence of the threshold voltage")
grid on;
```



Since the energies of the red and the yellow LEDs are small, the noise is much more severe when it comes to plotting.

Task 8:

```
U = Uth_arr(1:3)
```

```
U = 3×1
   -0.4363
   -0.4674
   -0.1873
```

```
f = Freq_arr(1:3)
```

```
f = 1×3
10^14 ×
    7.5113    6.4224    5.7468
```

```
linearFit = polyfit(f, U, 1);
planckSlope = linearFit(1)
```

```
planckSlope =
-1.2506e-15
```

```
planckIntercept = linearFit(2)
```

```
planckIntercept =
0.4568
```

```
e = -1.6e-19;
h = planckSlope * e
```

```
h =
2.0009e-34
```

```
W = planckIntercept * e
```

```
W =
-7.3081e-20
```

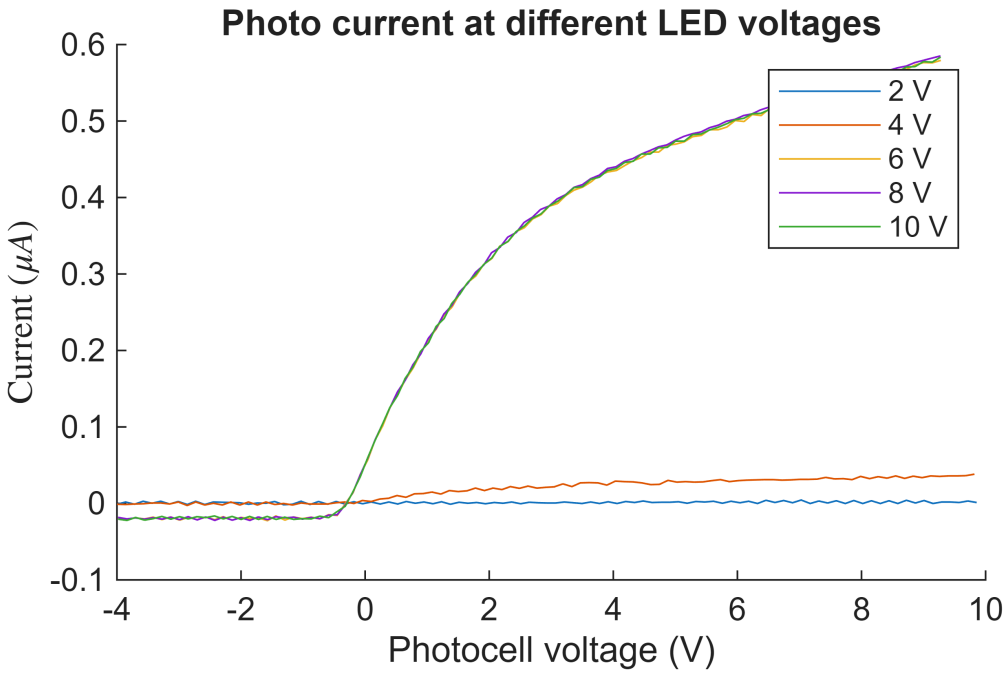
```
W_ev = W*6.2415e18
```

```
W_ev =
-0.4561
```

Task 9:

```
figure;
hold on;
data = readtable("task9_2.txt");
intensity = data{:, 2};
```

```
position = data{:, 1};
plot(position, intensity);
data = readtable("task9_4.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("task9_6.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("task9_8.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("task9_10.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
xlabel("Photocell voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("2 V", "4 V", "6 V","8 V", "10 V");
title("Photo current at different LED voltages")
```



Measurement 2

Task 10:

The aquired turn on voltages are:

UV LED: 2.3898 Volts

Blue LED: 2.1659 Volts

Green LED: 1.8716 Volts

Yellow LED: 1.5093 Volts

Red LED: 1.3679 Volts

```
Uto_arr = zeros(5,1);
```

UV:

```
figure;
hold on;
data = readtable("T10Uv.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position < 2);
constantPosition = position(position < 2);
constFit = polyfit(constantPosition,constantIntensity,1);
constSlope = constFit(1);
constIntercept = constFit(2);
x = linspace(-4,4,100);
y = constIntercept + x * constSlope;
plot(x(y > -.5 & y < 3),y(y > -.5 & y < 3));

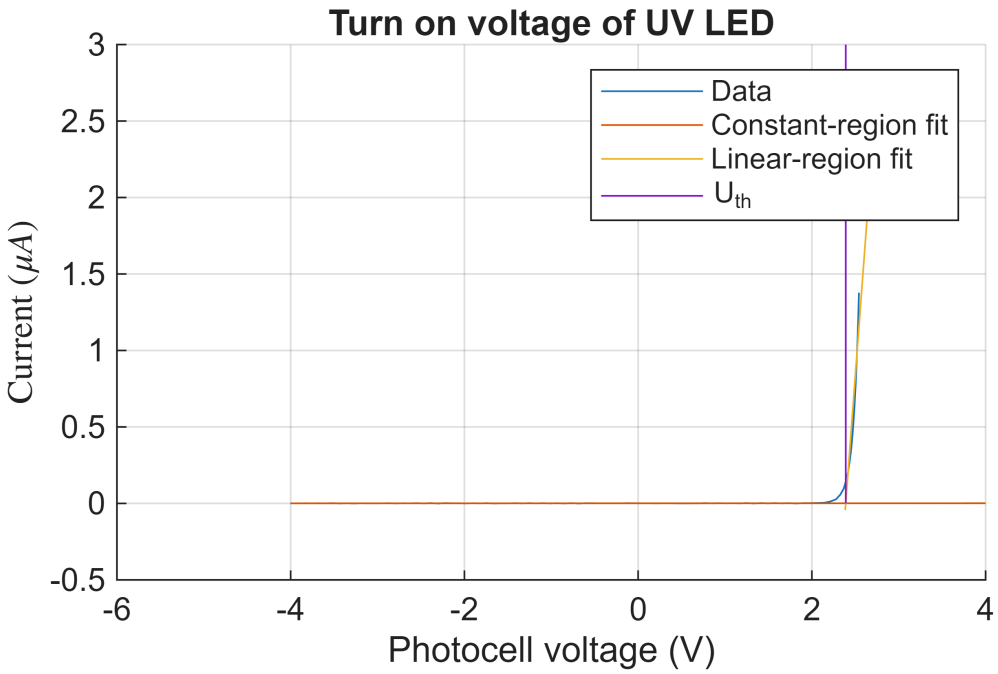
linIntensity = intensity(position > 2.36 & position < 4);
linPosition = position(position > 2.36 & position < 4);
linFit = polyfit(linPosition, linIntensity, 1);
linSlope = linFit(1);
linIntercept = linFit(2);
y = linIntercept + linSlope * x;
plot(x(y > -.5 & y < 3), y(y > -.5 & y < 3));

Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
2.3898
```

```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(0, 3,20))

xlabel("Driving voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{o}");
title("Turn on voltage of UV LED")
grid on;
```




```
Uto_arr(1) = Uth;
```

Blue:

```
figure;
hold on;
data = readtable("T10Blue.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position < 0);
constantPosition = position(position < 0);
constFit = polyfit(constantPosition,constantIntensity,1);
constSlope = constFit(1);
constIntercept = constFit(2);
x = linspace(-4,4,100);
y = constIntercept + x * constSlope;
plot(x(y > -.5 & y < 3),y(y > -.5 & y < 3));

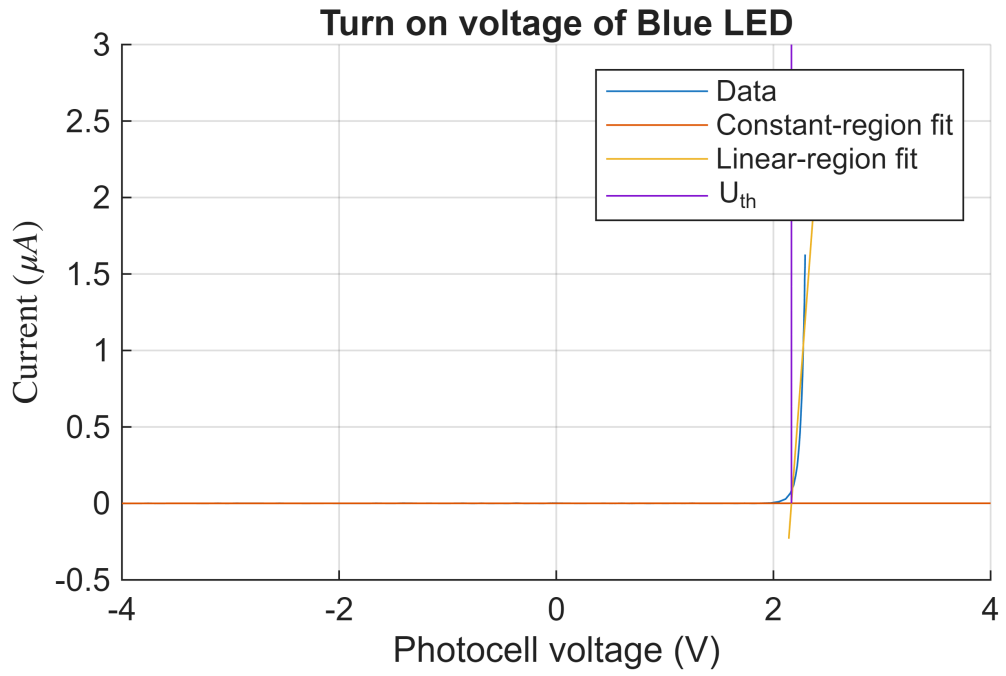
linIntensity = intensity(position > 2.1 & position < 4);
linPosition = position(position > 2.1 & position < 4);
linFit = polyfit(linPosition, linIntensity, 1);
linSlope = linFit(1);
linIntercept = linFit(2);
y = linIntercept + linSlope * x;
plot(x(y > -.5 & y < 3), y(y > -.5 & y < 3));
```

```
Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
2.1659
```

```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(0, 3,20))

xlabel("Driving voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{o}");
title("Turn on voltage of Blue LED")
grid on;
```




```
Uto_arr(2) = Uth;
```

Green:

```
figure;
hold on;
data = readtable("T10Green.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position < 0);
constantPosition = position(position < 0);
constFit = polyfit(constantPosition,constantIntensity,1);
constSlope = constFit(1);
constIntercept = constFit(2);
x = linspace(-4,4,100);
```

```
y = constIntercept + x * constSlope;
plot(x(y > -.5 & y < 3),y(y > -.5 & y < 3));

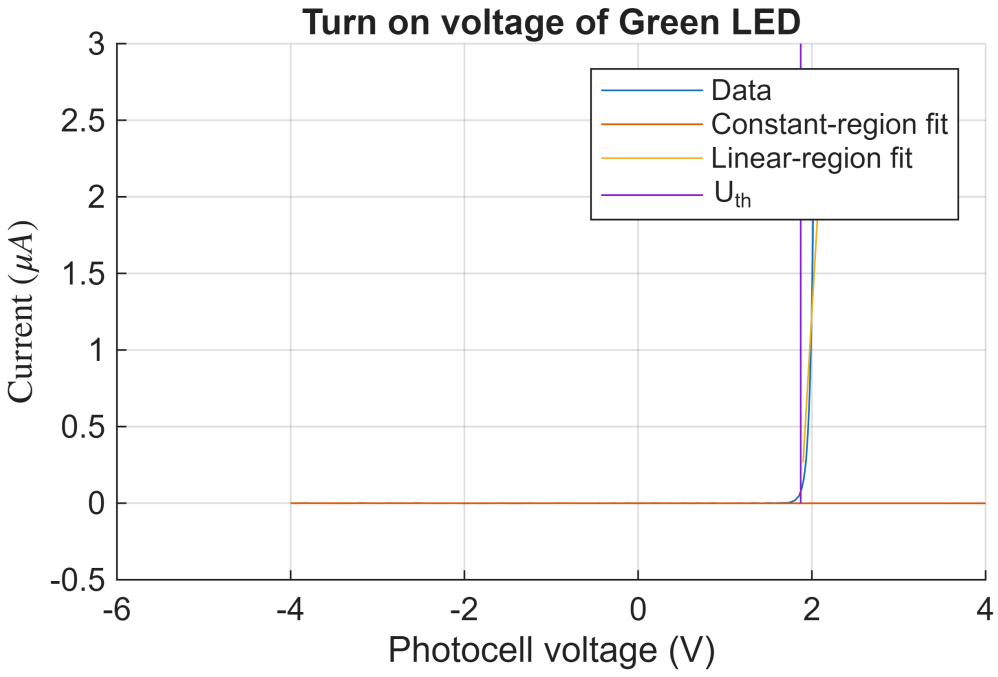
linIntensity = intensity(position > 1.8 & position < 4);
linPosition = position(position > 1.8 & position < 4);
linFit = polyfit(linPosition, linIntensity, 1);
linSlope = linFit(1);
linIntercept = linFit(2);
y = linIntercept + linSlope * x;
plot(x(y > -.5 & y < 3), y(y > -.5 & y < 3));

Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
1.8716
```

```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(0, 3,20))

xlabel("Driving voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{o}");
title("Turn on voltage of Green LED")
grid on;
```




```
Uto_arr(3) = Uth;
```

Yellow:

```
figure;
hold on;
```

```
data = readtable("T10Yellow.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position < 0);
constantPosition = position(position < 0);
constFit = polyfit(constantPosition,constantIntensity,1);
constSlope = constFit(1)
```

```
constSlope =
1.5931e-04
```

```
constIntercept = constFit(2)
```

```
constIntercept =
-0.0014
```

```
x = linspace(-4,4,100);
y = constIntercept + x * constSlope;
plot(x(y > -.5 & y < 3),y(y > -.5 & y < 3));

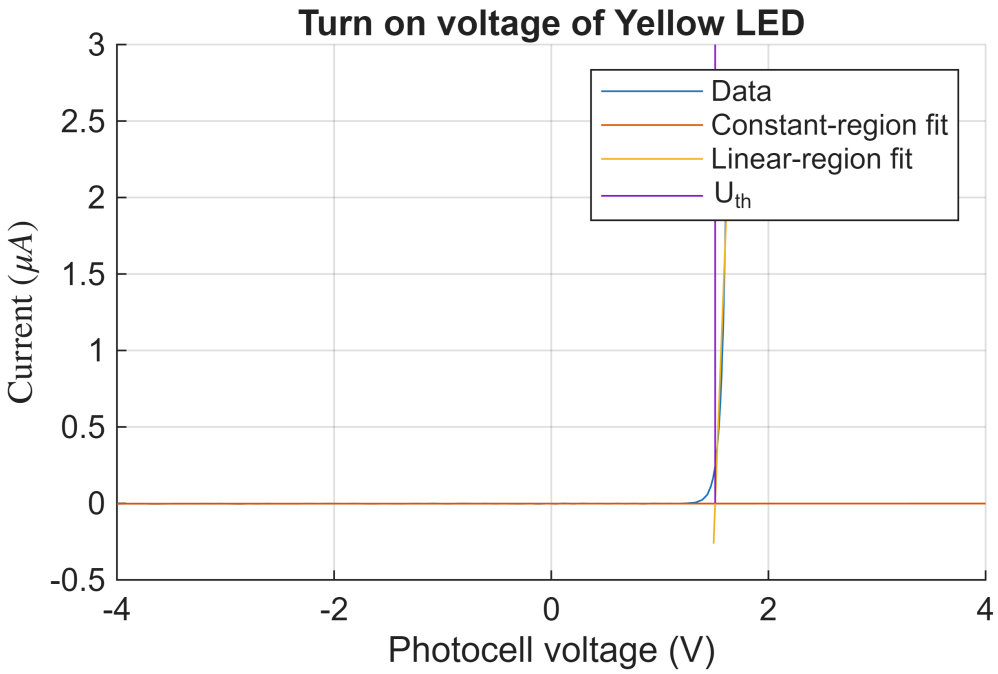
linIntensity = intensity(position > 1.49 & position < 4);
linPosition = position(position > 1.49 & position < 4);
linFit = polyfit(linPosition, linIntensity, 1);
linSlope = linFit(1);
linIntercept = linFit(2);
y = linIntercept + linSlope * x;
plot(x(y > -.5 & y < 3), y(y > -.5 & y < 3));

Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
1.5093
```

```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(0, 3,20))

xlabel("Driving voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{o}");
title("Turn on voltage of Yellow LED")
grid on;
```




```
Uto_arr(4) = Uth;
```

Red:

```
figure;
hold on;
data = readtable("T10Red.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);

constantIntensity = intensity(position < 0);
constantPosition = position(position < 0);
constFit = polyfit(constantPosition,constantIntensity,1);
constSlope = constFit(1)
```

```
constSlope =
3.3578e-04
```

```
constIntercept = constFit(2)
```

```
constIntercept =
-0.0020
```

```
x = linspace(-4,4,100);
y = constIntercept + x * constSlope;
plot(x(y > -.5 & y < 3),y(y > -.5 & y < 3));

linIntensity = intensity(position > 1.32 & position < 4);
linPosition = position(position > 1.32 & position < 4);
linFit = polyfit(linPosition, linIntensity, 1);
```

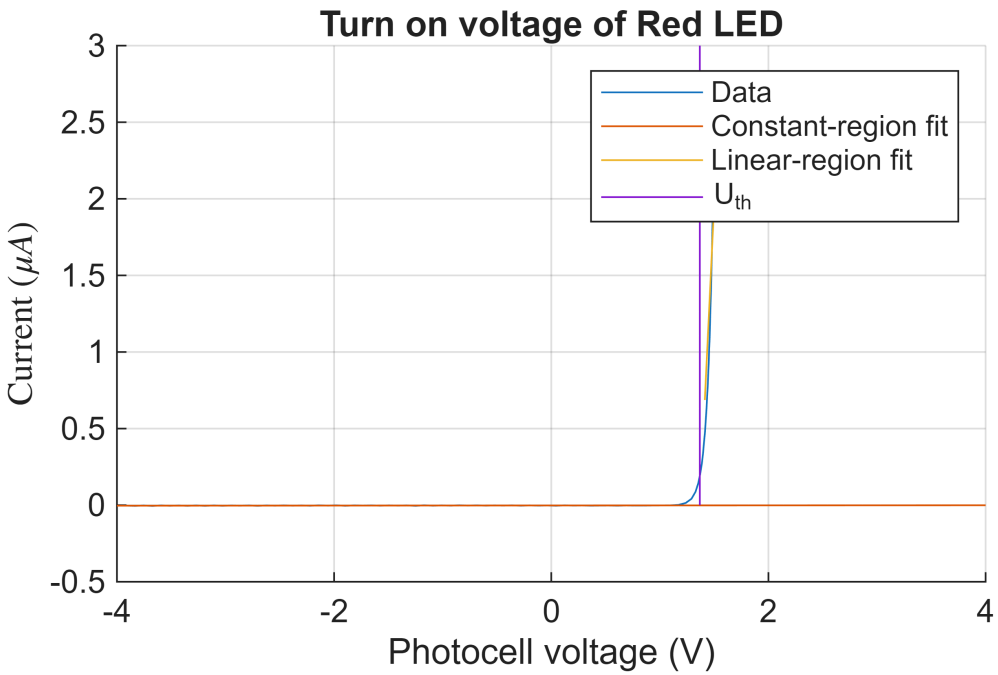
```
linSlope = linFit(1);
linIntercept = linFit(2);
y = linIntercept + linSlope * x;
plot(x(y > -.5 & y < 3), y(y > -.5 & y < 3));

Uth = (constIntercept - linIntercept) / (linSlope - constSlope)
```

```
Uth =
1.3679
```

```
Ith = linIntercept + linSlope*Uth;
plot(linspace(Uth,Uth,20),linspace(0, 3,20))

xlabel("Driving voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
legend("Data", "Constant-region fit", "Linear-region fit","U_{o}");
title("Turn on voltage of Red LED")
grid on;
```



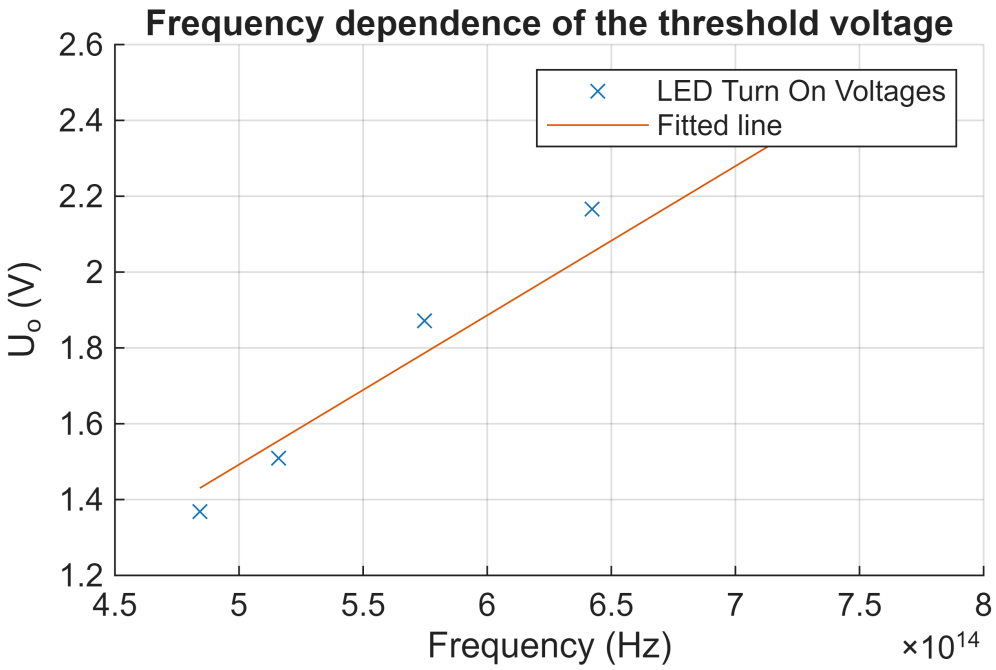

```
Uto_arr(5) = Uth;
```

Task 11:

```
figure;
hold on;
plot(Freq_arr,Uto_arr,'LineStyle','none','Marker','x')
ylabel("U_{o} (V)")
xlabel("Frequency (Hz)")
title("Frequency dependence of the threshold voltage")
grid on;
```

```
U = Uto_arr(1:5);
f = Freq_arr(1:5);

linearFit = polyfit(f, U, 1);
planckSlope = linearFit(1);
planckIntercept = linearFit(2);
plot(f,planckIntercept + planckSlope*f)
legend("LED Turn On Voltages", "Fitted line")
```




```
e = 1.6e-19;
h = planckSlope * e
```

```
h =
6.2964e-34
```

```
%W = planckIntercept * e

%W_ev = W*6.2415e18
```

Task 12:

The effect of the 1 $M\Omega$ resistor and the applied measureing voltage is that over a certain driving current the photocurrent bottoms out and can not be increased by increasing the driving current. To mitigate this we first search for the maximum driving current achiveable before saturating the measuring device, then we choose values between 2 mA and that maximum driving current.

```
% Calculating the drive voltage
Il = 2e-3;
R = 220;
```

```
Ud_min = (Il*R) + Uto_arr
```

```
Ud_min = 5×1
    2.8298
    2.6059
    2.3116
    1.9493
    1.8079
```

```
Il = 30e-3;
Ud_max = 5.34 %s (Il*R) + Uto_arr
```

```
Ud_max =
5.3400
```

```
Ud = linspace(Ud_min(2),Ud_max,10)
```

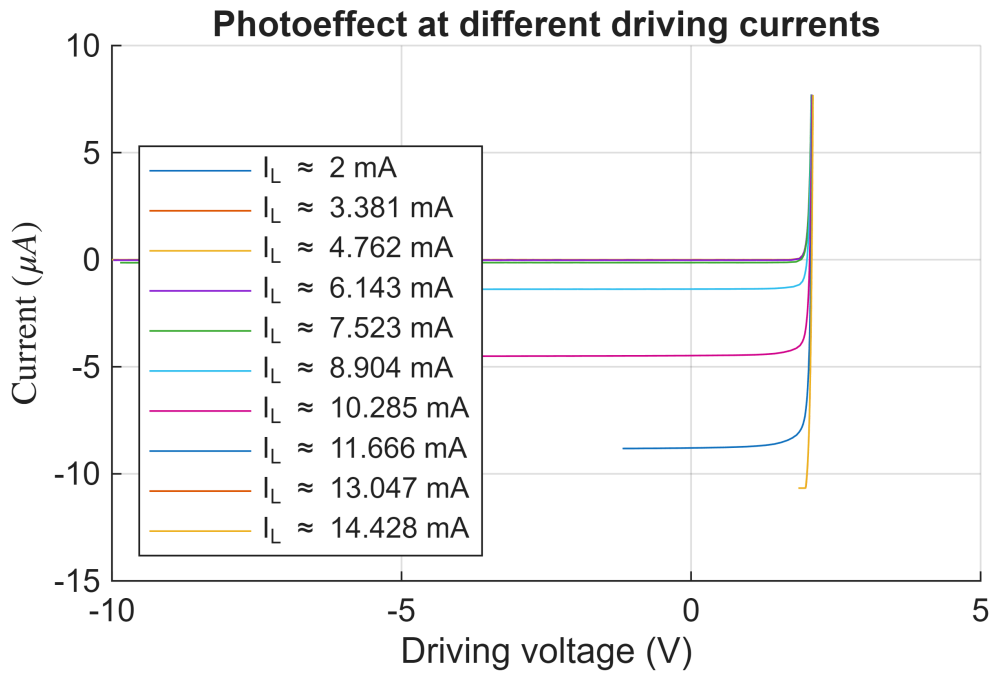
```
Ud = 1×10
    2.6059    2.9097    3.2135    3.5173    3.8211    4.1248    4.4286    4.7324 ···
```

```
figure;
hold on;
data = readtable("T12_1.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T12_2.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T12_3.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T12_4.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T12_5.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T12_6.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T12_7.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T12_8.txt");
```

```
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T12_9.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T12_10.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
Il = round((Ud - Uto_arr(2)) ./ R * 1e3, 3)
```

```
Il = 1×10
    2.0000    3.3810    4.7620    6.1430    7.5230    8.9040   10.2850   11.6660 ···
```

```
title("LED Photocell characteristics at different driving currents")
legend("I_L \approx {"+Il(1)+" mA}","I_L \approx {"+Il(2)+" mA}","I_L \approx
{"+Il(3) ...
    +" mA}","I_L \approx {"+Il(4)+" mA}","I_L \approx {"+Il(5)+" mA}","I_L \approx
{"+Il(6) ...
    +" mA}","I_L \approx {"+Il(7)+" mA}","I_L \approx {"+Il(8)+" mA}","I_L \approx
{"+Il(9)+" mA}", ...
    "I_L \approx {"+Il(10)+" mA}","Location","southwest")
xlabel("Driving voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
title("Photoeffect at different driving currents");
grid on;
```



Task 13:

```
Il = 10e-3;
R = 220;
Ud = (Il*R) + Uto_arr
```

```
Ud = 5×1
    4.5898
    4.3659
    4.0716
    3.7093
    3.5679
```

Driving current: 10 mA

Driving voltages:

UV: 4.5898

Blue: 4.3659

Green: 4.0716

Yellow: 3.7093

Red: 3.5679

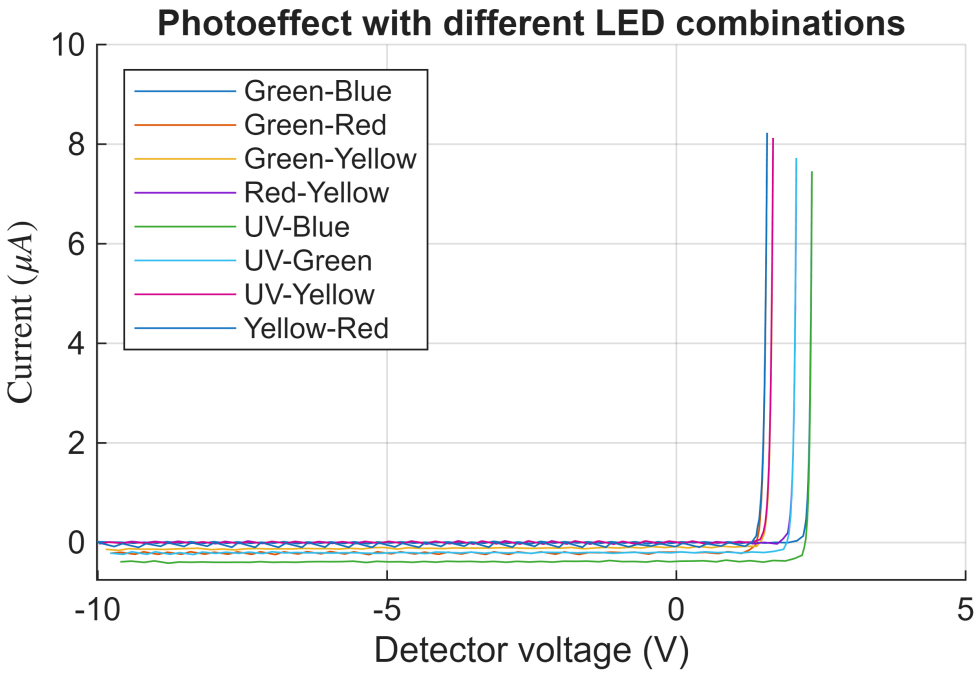
Tested combinations:

Driving LED	Detecting LED	Result
Yellow	Red	Small amount of photocurrent
Green	Blue	Not enough energy to excite the LED
Green	Red	Ok
Green	Yellow	Ok
UV	Blue	Ok
UV	Green	Ok
UV	Yellow	The photons produced by the UV LED have too much energy so they don't generate photocurrent
Red	Yellow	Not enough energy to excite the LED

On the legned the first LED is the driving LED, the second LED is the detector LED.

```
figure;
hold on;
data = readtable("T13GreenBlue.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T13GreenRed.txt");
```

```
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T13GreenYellow.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T13RedYellow.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T13UVBlue.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T13UVGreen.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T13UVYellow.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
data = readtable("T13YellowRed.txt");
intensity = data{:, 2};
position = data{:, 1};
plot(position, intensity);
xlabel("Detector voltage (V)");
ylabel("Current (\(\mu A)\)", "Interpreter","latex");
title("Photoeffect with different LED combinations")
legend("Green-Blue","Green-Red","Green-Yellow","Red-Yellow","UV-Blue","UV-
Green","UV-Yellow","Yellow-Red","Location","northwest")
ylim([-.75,10])
grid on;
```


Photoeffect with different LED combinations
Green-Blue
Green-Red
Green-Yellow
Red-Yellow
UV-Blue
UV-Green
UV-Yellow
Yellow-Red
Current (μA)
Detector voltage (V)
10
8
6
4
2
0
-10
-5
0
5